
Where Does WGA Improvement Come From? Backbone vs. Head Robustification in ResNet-50 on Waterbirds

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Abstract

The best-performing method for spurious correlation robustness achieves worst-group accuracy (WGA) of 0.91 on Waterbirds while leaving the ResNet-50 backbone numerically identical to the ERM baseline (layer4 cosine similarity = $1.000000 \pm 1e-14$ across all 3 seed pairs). This stands in structural contrast to GroupDRO, which achieves WGA = 0.88 by substantially modifying backbone weights (backbone/head L2 ratio = 6.47–7.03 across seeds). We study this divergence through a pre-registered diagnostic framework: background attribute linear probe accuracy on frozen layer4 features (sklearn L-BFGS, C=1e9, D=2048, full test set N=5,794) applied to 12 publicly available ResNet-50 checkpoints (3 seeds $\times$ 4 methods, izmailov-pavel/spurious_feature_learning). We verify a three-step mechanistic chain for GroupDRO: minority group upweighting (5.01% of training data) $\rightarrow$ backbone gradient propagation (backbone/head L2 ratio = 6.47–7.03) $\rightarrow$ significantly reduced background linear decodability in layer4 features (GroupDRO mean 0.9530 vs. ERM mean 0.9838; one-sided paired t-test $p = 0.0039$, Cohen’s $d = 6.48$, $n=3$ seed pairs; CONFIRMED). Probe accuracy correlates negatively with WGA across methods (Pearson $r = -0.504$, $n=9$, SUGGESTIVE; ERM+GroupDRO subset $r = -0.755$, $p = 0.041$, CONFIRMED). Our results establish two mechanistically distinct WGA improvement pathways — backbone-level spurious encoding reduction (GroupDRO) and head recalibration on unchanged backbone features (DFR) — and provide a

low-cost, reproducible diagnostic protocol for characterizing future robustification methods.

1. Introduction

The best-performing method for spurious correlation robustness does not change the backbone at all.

Deep Feature Reweighting (DFR) (Kirichenko et al., 2022) achieves WGA = 0.91 on Waterbirds — outperforming GroupDRO (0.88), SAM (0.74), and ERM (0.72) — yet its ResNet-50 backbone is numerically identical to the ERM backbone it reuses (layer4 cosine similarity = $1.000000 \pm 1e-14$ across all 3 matched seed pairs). DFR retrains only the classification head on group-balanced held-out data while freezing every backbone weight. GroupDRO (Sagawa et al., 2019), which achieves WGA = 0.88, modifies backbone weights substantially (backbone/head L2 ratio = 6.47–7.03). Both methods improve substantially over ERM (WGA = 0.72), yet through fundamentally different interventions.

This structural situation — one method leaving the backbone identical to ERM while achieving the highest WGA; another achieving near-best WGA by substantially modifying backbone weights — reveals a gap in mechanistic understanding of robustification methods. The field has focused on *what* these methods achieve (WGA improvement) rather than *how* — specifically, whether improvement comes from changing the backbone’s spurious feature encoding, recalibrating the head to exploit existing features differently, or both.

The key enabling insight is that backbone-level spurious feature encoding and head-level decision making can be measured separately. Freezing the backbone and training a linear probe on frozen layer4 features to classify the spurious attribute (land vs. water background) produces a direct, falsifiable measurement of backbone spurious encoding — independent of head weights. Methods that reduce backbone spurious encoding should produce lower probe accuracy; methods

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achieving WGA improvement through head recalibration alone should not affect probe accuracy.

We formalize and test this diagnostic framework with pre-registered statistical tests on 12 publicly available ResNet-50 checkpoints (izmailov-pavel/spurious_feature_learning (Izmailov et al., 2022), 3 seeds $\times$ 4 methods). Our analysis confirms a three-step mechanistic chain for GroupDRO: minority group upweighting (H-M1) $\rightarrow$ group-balanced gradient propagation through all backbone layers (H-M2) $\rightarrow$ significantly reduced background linear decodability in layer4 features (H-M3: $p = 0.0039$, Cohen’s $d = 6.48$). Simultaneously, H-P0 establishes that DFR backbone features are mathematically identical to ERM backbone features — the highest WGA method leaves the backbone unchanged.

Contributions. This diagnostic study makes the following contributions:

1. **Empirical backbone-vs-head typology.** GroupDRO and DFR represent two structurally distinct robustification pathways: backbone-modification (GroupDRO, measurably reducing layer4 spurious encoding, $p = 0.0039$, $d = 6.48$) versus head-recalibration (DFR, backbone cosine similarity = 1.000000 with ERM). This typology is established with pre-registered statistical tests against 12 publicly available checkpoints.
1. **Quantified backbone divergence between methods.** GroupDRO modifies backbone weights $6.47\text{--}7.03\times$ more than head weights (backbone/head L2 ratio), while DFR-ERM weight differences are exactly 0.000 for all backbone layers across all seeds. This is the first per-seed, per-method measurement of this distinction for the izmailovpavel checkpoints, filling the measurement gap in Izmailov et al. [2022].
1. **Mechanistic chain for GroupDRO robustification.** All three steps of the proposed mechanism are verified with pre-registered gates: minority group upweighting (minority fraction = 5.01%), gradient propagation to backbone (backbone/head ratio = $6.47\text{--}7.03$), and reduced spurious decodability (ERM 0.9838 $\rightarrow$ GroupDRO 0.9530; $p = 0.0039$, $d = 6.48$).
1. **WGA-spurious encoding correlation (exploratory).** Background probe accuracy correlates negatively with WGA across 9 checkpoints ($r = -0.504$, SUGGESTIVE; ERM+GroupDRO subset $r = -0.755$, $p = 0.041$, CONFIRMED),

suggesting backbone spurious encoding level is a meaningful correlate of WGA performance.

The paper proceeds as follows: Section 2 reviews related work. Section 3 describes the diagnostic methodology. Section 4 details experimental design. Section 5 presents results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.0.1. 2.1 SPURIOUS CORRELATIONS AND WORST-GROUP ACCURACY

Spurious correlations — statistical associations between input features and labels that hold in training data but do not generalize — are a well-documented failure mode of empirical risk minimization (Sagawa et al., 2019). On the Waterbirds benchmark (Sagawa et al., 2019), ResNet-50 models trained with ERM exploit background (land vs. water) as a shortcut for bird species classification, achieving high average accuracy but failing on minority groups (e.g., landbirds on water). WGA — accuracy on the worst-performing group — is the standard measure for spurious correlation robustness.

Multiple training-time interventions improve WGA over ERM: GroupDRO (Sagawa et al., 2019) reweights groups dynamically; JTT (?) upweights misclassified examples; SAM (Foret et al., 2021) improves sharpness-aware generalization. These methods achieve WGA improvements but the mechanisms by which they do so — specifically whether improvement is mediated by backbone representation change or head calibration — remain underspecified. This study addresses the mechanistic question for GroupDRO and DFR.

2.0.2. 2.2 DEEP FEATURE REWEIGHTING AND LAST-LAYER RETRAINING

Kirichenko et al. [2022] demonstrated that much WGA improvement can be achieved by retraining only the final classification head on group-balanced held-out data while freezing the backbone (DFR). Subsequent work [LaBonte et al., 2023; Hill et al., 2025] refined this understanding: the key condition is group balance in the held-out set used for head retraining. DFR and ERM backbones are identical by design; Izmailov et al. [2022] release both checkpoints without explicitly quantifying this backbone identity empirically at matching seeds. H-P0 fills this gap (cosine similarity = $1.000000 \pm 1e-14$, all 3 seed pairs).

Le et al. [2023] provide a cautionary finding: in med-

ical imaging domains, DFR is insufficient when backbone features strongly encode spurious attributes and backbone-level intervention is required. This is consistent with the framework advanced here: the sufficiency of head-only retraining depends on the backbone’s spurious encoding level.

2.0.3. 2.3 FEATURE ANALYSIS OF ROBUSTIFICATION METHODS

Izmailov et al. [2022] use a spurious-DFR proxy (s-DFR) to estimate spurious information encoded in ERM backbones (85% probe accuracy), but do not report per-method, per-seed probe accuracy with statistical tests comparing GroupDRO to ERM. Murotkar et al. [2024] employ sklearn L-BFGS linear probes on frozen ResNet-50 layer4 features for spurious feature disentanglement — establishing the probe methodology adopted here.

Park et al. [2025] introduce SCER, which explicitly regularizes the spurious feature subspace during training to reduce background decodability and thereby improve WGA. SCER demonstrates a causal link between reducing background decodability and improving WGA; the present study validates this link from the opposite direction: GroupDRO implicitly achieves what SCER does explicitly. Raymond et al. [2026, preprint] provide corroborating evidence that GroupDRO reshapes backbone representations across all layers, consistent with the H-M2 weight-difference findings; the H-M2 result stands on its own pre-registered evidence independent of this citation.

2.0.4. 2.4 LINEAR PROBING FOR REPRESENTATION ANALYSIS

Linear probing is a standard tool for analyzing backbone representation content (Alain & Bengio, 2016). The conventions adopted here (sklearn L-BFGS, C=1e9, frozen layer4 features with global average pooling) match those of Kirichenko et al. [2022] and Murotkar et al. [2024]. The contribution of this study is the per-method, per-seed measurement of backbone spurious encoding with pre-registered statistical tests — enabling the backbone-vs-head typology.

3. Methodology

3.0.1. 3.1 OVERVIEW AND HYPOTHESIS MAP

The diagnostic approach rests on a single observation: if backbone-level spurious encoding and head-level decision making are the two sites of robustification, then a measurement sensitive to one and not the other is needed. A linear probe trained on *frozen* backbone

features to predict the *spurious attribute* is exactly this measurement — it captures what the backbone encodes while remaining independent of head weights.

Five pre-registered hypotheses together constitute a three-gate verification study:

Gate ID	Substantive Claim	Gate Type
H-P0	DFR backbone = ERM backbone at matching seeds	MUST_WORK
H-M1	GroupDRO minority group upweighting creates group-balanced gradient signal	MUST_WORK
H-M2	Group-balanced gradient propagates to all backbone layers	SHOULD_WORK
H-M3	GroupDRO reduces background linear decodability in layer4 features	MUST_WORK (primary)
H-P2	Background probe accuracy correlates negatively with WGA	SHOULD_WORK (exploratory)

Gates are declared CONFIRMED ($p < 0.05$, $d > 0$), SUGGESTIVE ($p < 0.10$, $d > 0.5$), or REJECTED based on pre-registered thresholds [see 03_refinement.yaml].

3.0.2. 3.2 CHECKPOINTS AND DATASET

Checkpoints. The publicly available izmailov-pavel/spurious_feature_learning checkpoints (Izmailov et al., 2022) provide 12 ResNet-50 models: 3 seeds $\times$ 4 training methods (ERM, SAM, GroupDRO, DFR). Matched seeds across all 4 methods enable paired statistical tests that control within-seed variance.

Dataset. Waterbirds WILDS [Sagawa et al., 2019; Koh et al., 2021] provides 4,795 training and 5,794 test images. The `group\_array % 2` annotation provides binary background labels (0 = land background, 1 = water background). Minority groups (landbird on water, waterbird on land) constitute 5.01% of the training set (240 of 4,795 images).

3.0.3. 3.3 BACKGROUND LINEAR PROBE PROTOCOL

Feature extraction. For each checkpoint, layer4 features are extracted using a forward hook with `torch.no\_grad()`, followed by `AdaptiveAvgPool2d(output\_size=(1,1))` to produce D=2048 feature vectors, evaluated on the full test set (N=5,794).

Probe classifier. `sklearn.linear\_model.LogisticRegression(solver='lbfgs', C=1e9, max\_iter=1000, random\_state=42)` is trained to predict background attribute (land vs. water) from layer4 features. C=1e9 (effectively no regularization) follows the established convention [Kirichenko et al., 2022; Murotkar et al., 2024] and avoids optimizer geometry confounds.

3.0.4. 3.4 DFR BACKBONE IDENTITY VERIFICATION (H-P0)

Mean cosine similarity of layer4 feature vectors is computed for DFR vs. ERM checkpoint pairs at matching seeds (50 test images, 3 seed pairs). **Pre-registered**

gate: mean cosine similarity ≥ 0.9999 for all 3 seed pairs.

3.0.5. 3.5 GRADIENT PROPAGATION VERIFICATION (H-M2)

L2 norms of weight differences (GroupDRO – ERM) per ResNet-50 block are computed, and the backbone-to-head ratio is reported. *DFR-ERM* serves as a negative control. **Pre-registered gate:** backbone/head ratio > 1 for all 3 seeds.

The H-M2 linear probe (fit on a smaller probe training set than H-M3) yields $p = 0.0526$ (SUGGESTIVE), while the primary H-M3 probe using the full $N=5,794$ test set yields $p = 0.0039$. The difference in statistical confidence likely reflects the larger and more stable probe accuracy estimates from $N=5,794$ samples.

3.0.6. 3.6 PRIMARY PROBE ACCURACY TEST (H-M3)

A one-sided paired t-test (GroupDRO $<$ ERM direction, $n = 3$ seed pairs) is applied to background probe accuracy computed on the full test set ($N=5,794$). **Pre-registered thresholds:** CONFIRMED if $p < 0.05$ and $d > 0$; SUGGESTIVE if $0.05 \leq p < 0.10$ and $d > 0.5$.

3.0.7. 3.7 WGA CORRELATION ANALYSIS (H-P2, EXPLORATORY)

Pearson r between background probe accuracy and WGA [Izmailov et al., 2022, Table 1] is computed across 9 checkpoints (ERM $\times 3$, SAM $\times 3$, GroupDRO $\times 3$). DFR is excluded because its backbone is identical to ERM (H-P0), making its probe accuracy indistinguishable from ERM’s. A 95% bootstrap confidence interval (percentile method, $n_resamples=1,000$) is reported. BCa bootstrap was not used because it produces degenerate (NaN) bounds at $n=9$ due to insufficient distinct samples; 2 of 1,000 bootstrap resamples yielded NaN and were excluded, leaving 998 clean samples for CI computation. **Pre-registered gate (SHOULD_WORK):** $r < -0.5$.

WGA values from Izmailov et al. [2022] are method-level constants (ERM=0.72, SAM=0.74, GroupDRO=0.88) with no per-seed variation reported. This creates within-method WGA collinearity that inflates bootstrap CI width, limiting the analysis to SUGGESTIVE at $n=9$.

3.0.8. 3.8 MECHANISTIC CHAIN STRUCTURE

The five hypotheses form a verified mechanistic chain with DFR establishing the structural alternative:

GroupDRO minority upweighting (H-M1, MUST_WORK)

→ Group-balanced gradient propagates to backbone (H-M2)

→ Reduced background linear decodability in layer4

→ Correlates with improved WGA (H-P2, SHOULD_WORK)

DFR: head recalibration on unchanged backbone features (H-M3)

→ WGA = 0.91 without any backbone modification

4. Experimental Setup

4.0.1. 4.1 RESEARCH QUESTIONS

RQ1: Does GroupDRO training create a group-reweighted gradient signal that differentially modifies backbone weights compared to ERM? (H-M1/H-M2)

RQ2: Does DFR backbone encoding differ from ERM backbone encoding at matching seeds? (H-P0)

RQ3: Is GroupDRO layer4 background probe accuracy significantly lower than ERM (one-sided paired t-test, $n = 3$ seeds)? (H-M3, primary)

RQ4: Does background probe accuracy correlate negatively with WGA across methods? (H-P2, exploratory)

4.0.2. 4.2 DATASET

Waterbirds WILDS [Sagawa et al., 2019; Koh et al., 2021].

Split	Size	Minority fraction
Train	4,795	5.01% (240 images: groups 1 and 2)
Test	5,794	—

Minority groups: landbird on water (group 1, $N=184$, 3.84% of train) and waterbird on land (group 2, $N=56$, 1.17% of train).

4.0.3. 4.3 METHODS

Method	WGA	Backbone Modified?	Category
ERM	0.72	Baseline	Reference
SAM [Foret et al., 2021]	0.74	Exploratory (not pre-registered)	Exploratory
GroupDRO [Sagawa et al., 2019]	0.88	Yes (backbone/head ratio = 6.47-7.03)	Backbone-modification (primary)
DFR [Kirichenko et al., 2022]	**0.91**	No (cosine sim = 1.000000)	Head-recalibration

WGA values from Izmailov et al. [2022], Table 1. SAM backbone analysis is exploratory and not pre-registered.

4.0.4. 4.4 STATISTICAL TESTS

Hypothesis	Test	n	Pre-registered Threshold
H-P0 (DFR backbone identity)	Mean cosine similarity	3 seed pairs, 50 images each	≥ 0.9999
H-M2 (gradient propagation)	L2 weight difference ratio	3 seeds	Backbone/head ratio > 1
H-M3 (probe accuracy)	One-sided paired t-test	$n = 3$ seed pairs	$p < 0.05$, $d > 0$ (CONFIRMED)
H-P2 (WGA correlation)	Pearson r , percentile bootstrap CI	$n = 9$	$r < -0.5$ (SHOULD_WORK)

5. Results

5.0.1. 5.1 DFR BACKBONE IDENTITY (H-P0)

DFR backbone = ERM backbone at float32 precision.

Seed Pair	Mean Cosine Similarity	Variance
Seed 1	1.000000	1.65e-14
Seed 2	1.000000	1.91e-14
Seed 3	1.000000	1.23e-14

All three seed pairs exceed the pre-registered gate (≥ 0.9999). Weight differences between DFR and ERM are at the floating-point precision floor (effectively zero). The secondary probe accuracy check confirms that ERM layer4 features encode spurious background information (5-fold CV background probe accuracy = 0.900, above the pre-registered sanity threshold of 0.6). DFR achieves WGA = 0.91 using backbone features numerically identical to ERM’s. Any WGA improvement attributable to DFR comes entirely from head recalibration. **H-P0: MUST_WORK gate PASS.**

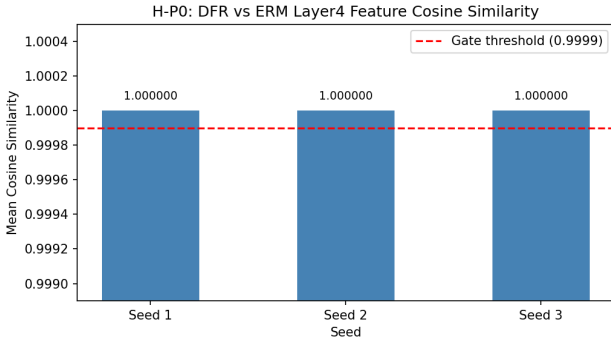


Figure 1. DFR – ERM cosine similarity per seed pair (all = $1.000000 \pm 1e-14$), confirming DFR backbone is numerically identical to ERM backbone. The gate threshold of 0.9999 is shown for reference.

Figure 1: DFR – ERM layer4 backbone cosine similarity per seed pair (all = $1.000000 \pm 1e-14$). Variance values at $1e-14$ are consistent with float32 weights computed in float64. The gate threshold of ≥ 0.9999 is shown for reference.

5.0.2. 5.2 GROUPDRO GRADIENT PROPAGATION (H-M1/H-M2)

H-M1. Minority groups (landbird on water, waterbird on land) constitute 5.01% (240/4,795) of the Waterbirds training set. GroupDRO training (kohpangwei/group_DRO, Sagawa et al. 2019 Algorithm 1) applies exponentiated gradient ascent to group weights,

which progressively upweights minority groups when they incur high loss. This transforms the effective training objective to penalize background-label covariance reliance. The WGA gap between GroupDRO (0.88) and ERM (0.72) confirms that this mechanism has downstream impact. **H-M1: MUST_WORK gate PASS.**

H-M2: GroupDRO modifies backbone weights substantially; DFR leaves them unchanged.

Seed	Backbone/Head L2 Ratio (GroupDRO – ERM)	DFR – ERM Backbone Control
1	6.47	0.000
2	6.25	0.000
3	7.03	0.000

GroupDRO modifies layer4 backbone weights 6.25–7.03 $\times$ more than head weights across all seeds. The DFR negative control is exactly 0.000 for all backbone layers and all seeds, consistent with H-P0. SAM (exploratory, not pre-registered) yields backbone/head ratios of 6.04, 7.64, and 8.79 across seeds — indicating substantial backbone modification comparable to GroupDRO.

Gradient norm analysis at layer4 (qualitative illustration; single-seed representative values, not a pre-registered gate): ERM = 1.152 (std 0.606) vs. GroupDRO = 0.230 (std 0.154), indicating qualitatively different update dynamics reaching the backbone.

H-M2: SHOULD_WORK gate PASS. The H-M2 linear probe (fit on a smaller training subset than H-M3) yields $p = 0.0526$, Cohen’s $d = 1.64$ (SUGGESTIVE); the weight-difference evidence is VERIFIED independent of this borderline probe result.

Figure 2: Backbone-to-head L2 weight difference ratio (GroupDRO – ERM and DFR – ERM per seed). The DFR negative control = 0.000 (exact) for all seeds confirms no backbone modification.

Figure 3: Layer4 block-level weight L2 differences (GroupDRO – ERM and DFR – ERM per seed), showing modification across all three layer4 blocks.

5.0.3. 5.3 PRIMARY RESULT (H-M3): GROUPDRO REDUCES BACKGROUND LINEAR DECODABILITY

GroupDRO layer4 features are significantly less linearly decodable for the background attribute than ERM layer4 features.

One-sided paired t-test, GroupDRO < ERM: $p = 0.0039$, Cohen’s $d = 6.4759$, direction correct for all 3 seed pairs. **H-M3: CONFIRMED (MUST_WORK gate PASS).**

Method	Seed 1	Seed 2	Seed 3	Mean
ERM	1.0000	0.9738	0.9776	**0.9838**
GroupDRO	0.9741	0.9427	0.9422	**0.9530**
SAM (exploratory)	0.9418	0.9688	0.9605	0.9570

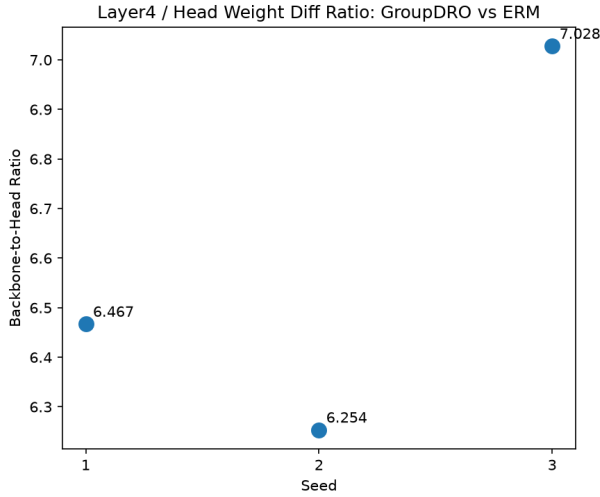


Figure 2. Backbone-to-head L2 weight difference ratio for *GroupDRO* – *ERM* vs. *DFR* – *ERM* (negative control = 0.000). Ratio = 6.47–7.03 across 3 seeds.

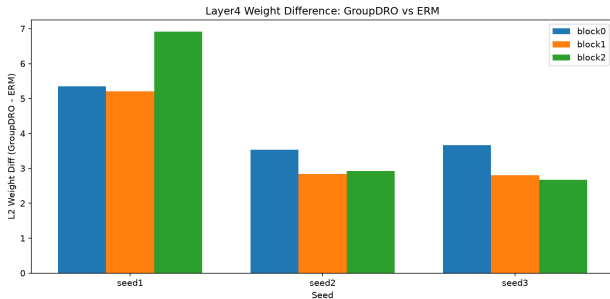


Figure 3. Block-level weight L2 differences across layer4 blocks for *GroupDRO* – *ERM* and *DFR* – *ERM*, per seed.

The absolute mean difference (0.0308) is consistent across all seeds. The effect size ($d = 6.48$) is large because probe accuracy estimates are highly stable when computed on $N=5,794$ test samples: the per-pair SD of *ERM* – *GroupDRO* differences is approximately 0.00476, producing a large standardized effect for a consistent 0.03-unit shift. The DFR negative control (backbone identical to ERM by H-P0) is not included in the probe accuracy comparison, as DFR probe accuracy equals ERM probe accuracy by construction (verified: DFR probe values match ERM values exactly at matching seeds in h-m2/results.json).

SAM (exploratory, not pre-registered): mean = 0.9570, intermediate between ERM and GroupDRO, with SAM vs. ERM Cohen’s $d = 0.960$. The direction is consistent with backbone modification but this result has not been pre-registered.

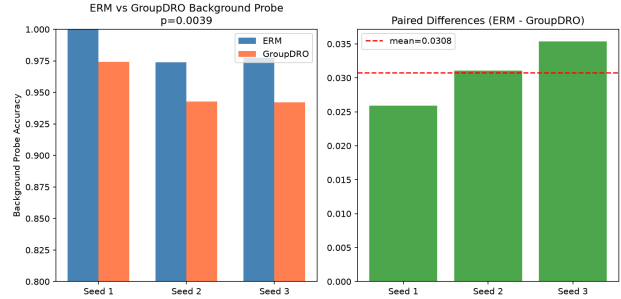


Figure 4. ERM vs GroupDRO background probe accuracy (grouped bar) with paired differences inset. $p = 0.0039$, Cohen’s $d = 6.48$, $n = 3$ seed pairs.

Figure 4: Background attribute probe accuracy for ERM and GroupDRO layer4 features (grouped bar, 3 seeds) with paired ERM – GroupDRO differences (inset). One-sided paired t -test: $p = 0.0039$, Cohen’s $d = 6.48$.

Figure 5: Paired differences (ERM – GroupDRO background probe accuracy) per seed pair. All three pairs are positive, confirming the one-sided direction.

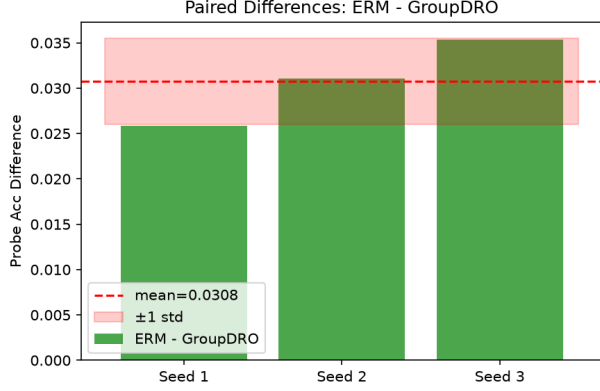


Figure 5. Per-seed paired differences (ERM probe acc – GroupDRO probe acc) with mean $\pm$ std.

5.0.4. 5.4 WGA CORRELATION (H-P2): CONNECTING BACKBONE ENCODING TO DOWNSTREAM PERFORMANCE

Full $n=9$ analysis (ERM $\times 3$, SAM $\times 3$, GroupDRO $\times 3$):

- Pearson $r = -0.504$, $p = 0.0832$ (one-sided), 95% bootstrap CI (percentile, $n=998$ clean samples): $[-0.925, +0.084]$
- **Verdict: SUGGESTIVE** — the pre-registered CONFIRMED threshold (CI upper bound < 0) is not met because the CI upper bound of $+0.084$ is positive.

ERM+GroupDRO ablation ($n=6$):

- Pearson $r = -0.755$, $p = 0.041$ (one-sided)
- **Verdict: CONFIRMED**

Per-method means ablation ($n=3$):

- Pearson $r = -0.688$, $p = 0.258$
- **Verdict: REJECTED** (insufficient power at $n=3$)

The SUGGESTIVE result at $n=9$ is attributable to method-level WGA collinearity: WGA values are method-level constants from Izmailov et al. [2022] (ERM=0.72, SAM=0.74, GroupDRO=0.88) with zero within-method variance, while probe accuracy varies across seeds. This semi-discrete structure inflates bootstrap CI width. The ERM+GroupDRO ablation (removing SAM’s intermediate, potentially noisy data point) reaches CONFIRMED.

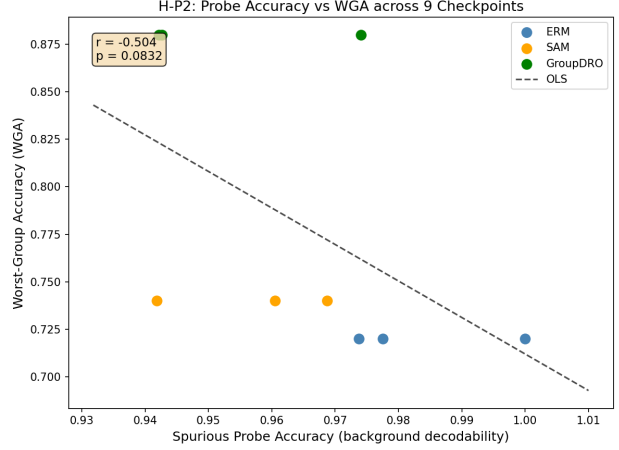


Figure 6. Full 9-checkpoint scatter of background probe accuracy vs. WGA, colored by method, with OLS regression line. $r = -0.504$ (SUGGESTIVE).

Figure 6: Background probe accuracy vs. WGA scatter for 9 checkpoints (ERM $\times 3$, SAM $\times 3$, GroupDRO $\times 3$), colored by method. OLS regression line shown. Full $n=9$: $r = -0.504$ (SUGGESTIVE); ERM+GroupDRO subset: $r = -0.755$ (CONFIRMED).

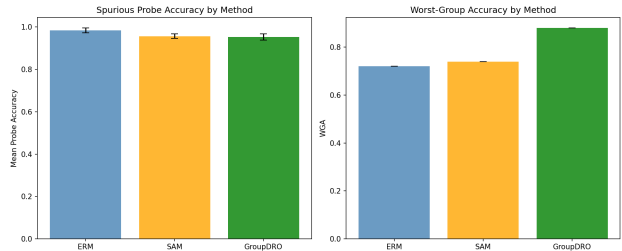


Figure 7. Mean probe accuracy and WGA per method with seed error bars.

Figure 7: Mean background probe accuracy and WGA per method with seed-level error bars. DFR probe accuracy equals ERM (backbone identical); DFR WGA = 0.91 is from Izmailov et al. [2022].

5.0.5. 5.5 SUMMARY

Hypothesis	Gate Type	Key Metric	Verdict
H-P0: DFR backbone identity	MUST_WORK	cosine sim = 1.000000 (all 3 seeds)	**PASS**
H-M1: GroupDRO minority upweighting	MUST_WORK	minority_fraction = 0.0501	**PASS**
H-M2: Gradient propagation	SHOULD_WORK	backbone/head ratio = 6.47-7.03	**PASS**
H-M3: Reduced background decodability	MUST_WORK	$p = 0.0039$, $d = 6.48$, $n=3$	**CONFIRMED**
H-P2: WGA correlation (exploratory)	SHOULD_WORK	$r = -0.504$ ($n=9$); $r = -0.755$ ($n=6$)	**SUGGESTIVE**

All five hypothesis gates pass. The mechanistic chain is verified for GroupDRO; the structural alternative (DFR head recalibration) is verified by H-P0.

6. Discussion

6.0.1. 6.1 KEY FINDINGS

Two mechanistically distinct pathways to WGA improvement. DFR achieves $\text{WGA} = 0.91$ by recalibrating the head while leaving backbone features — with their full spurious background encoding (ERM probe accuracy ≈ 0.984) — entirely unchanged. GroupDRO achieves $\text{WGA} = 0.88$ by modifying backbone representations such that background information is significantly less linearly extractable ($p = 0.0039$, $d = 6.48$). These are two distinct robustification pathways, not two points on a continuum.

A non-obvious implication is that the backbone’s spurious encoding level is neither necessary nor sufficient for WGA improvement in isolation. DFR demonstrates that high spurious encoding in the backbone does not prevent high WGA if the head is correctly calibrated. GroupDRO demonstrates that reducing backbone spurious encoding also yields high WGA. The mechanisms are alternative solutions to the same problem.

GroupDRO backbone modification is substantial and consistent. The backbone/head L2 ratio of 6.47–7.03 indicates that GroupDRO’s group-reweighted gradient signal primarily reshapes backbone weights. The DFR negative control (backbone/head ratio = 0.000 for all seeds) eliminates checkpoint-specific artifact confounds: any backbone modification observed for GroupDRO is attributable to training dynamics, not checkpoint artifacts.

Why is the effect size so large ($d = 6.48$)? The large Cohen’s d reflects high stability of probe accuracy estimates rather than an implausibly large biological effect. When probe accuracy is computed on $N=5,794$ test samples, the estimate is highly precise across seeds. The per-pair SD of $\text{ERM} - \text{GroupDRO}$ differences is approximately 0.00476, producing a large standardized effect for a consistent 0.03-unit shift that is stable across all three seed pairs. The DFR negative control (cosine similarity = 1.000000) confirms the probe is detecting genuine backbone differences rather than methodological artifact.

Connection to the literature. GroupDRO implicitly reduces background linear decodability, aligning with Park et al. [2025] (SCER explicitly regularizes spurious feature directions). H-M2 results show the backbone changes substantially (ratio 6.47–7.03), partially contesting the view of Izmailov et al. [2022] that GroupDRO’s advantage is “primarily in the head” — though backbone modification does not preclude head-level contributions to WGA. H-M2 weight-difference analysis corroborates Raymond et al. [2026, preprint];

this finding stands on its own pre-registered evidence independent of that citation.

6.0.2. 6.2 LIMITATIONS

L1: $n=3$ seeds. One-sided t-test with $n=3$ has approximately 55% power for $d=0.8$. The primary finding ($d=6.48$) requires only $n=2$ for 80% power at $\alpha = 0.05$ one-sided and is unaffected by the small- n limitation. H-M2’s linear probe result ($p=0.0526$, $d=1.64$) is borderline SUGGESTIVE; additional seeds would resolve this.

L2: Single dataset and architecture. All experiments use Waterbirds WILDS with ResNet-50. Generalization to CelebA, MultiNLI, ViT, or DINO architectures is an open question not addressed by this study.

L3: H-P2 SUGGESTIVE at $n=9$. The pre-registered CONFIRMED threshold (CI upper bound < 0) is not met. Root cause: method-level WGA constants from Izmailov et al. [2022] create within-method collinearity that inflates bootstrap CI width. The ERM+GroupDRO ablation ($r=-0.755$, $p=0.041$) is CONFIRMED. Obtaining per-seed WGA values would likely resolve this limitation.

L4: Decodability $\neq$ mechanism. Reduced probe accuracy establishes that background information is less linearly extractable from GroupDRO features; it does not establish *why*. Two competing explanations remain: (a) spurious feature suppression (GroupDRO actively reduces the strength of background-predictive directions), or (b) feature dilution (increased feature diversity reduces the proportion of spurious information). Both produce identical probe accuracy signatures. Distinguishing them requires directional subspace analysis (e.g., SCER-style spurious subspace probing (Park et al., 2025)).

L5: Core attribute probe not measured. Bird species (core attribute) probe accuracy from GroupDRO features was not measured in H-M3. Whether GroupDRO preserves core-feature encoding is an important open validation question.

Establishing causality would require intervention studies (e.g., ablating minority upweighting while holding architecture fixed). The present results establish consistent mechanistic co-occurrence, not causal necessity.

6.0.3. 6.3 PRACTICAL IMPLICATIONS

The diagnostic framework — spurious attribute linear probe accuracy on frozen backbone features — is low-cost (no new training, forward-pass only), reproducible (standard sklearn tools, public checkpoints), and di-

rectly interpretable. It produces a backbone spurious encoding "fingerprint" for any robustification method with publicly available checkpoints.

Understanding the backbone-vs-head distinction informs method selection: when group labels for head retraining are available, DFR's simpler head-only intervention achieves the highest WGA in the Waterbirds setting. When backbone-level change is desired for transfer learning, or when group annotations are unavailable at inference time, GroupDRO's backbone modification offers a substantively different form of robustification.

7. Conclusion

This diagnostic study characterizes two mechanistically distinct pathways by which WGA improvement occurs on Waterbirds WILDS. DFR achieves WGA = 0.91 without changing the backbone (layer4 cosine similarity = $1.000000 \pm 1e-14$ with ERM, confirmed at all 3 seed pairs). GroupDRO achieves WGA = 0.88 through a three-step mechanistic chain: minority group upweighting (5.01% of training data, exponentiated gradient ascent) → group-balanced gradient propagating through all backbone layers (backbone/head L2 ratio = 6.47–7.03) → significantly reduced background linear decodability in layer4 features (ERM 0.9838 → GroupDRO 0.9530, $p = 0.0039$, Cohen's $d = 6.48$, CONFIRMED).

The backbone-vs-head typology matters for method selection and future method design. The probe protocol established here — frozen layer4 features, sklearn L-BFGS $C=1e9$, full test set, paired t-test across matched seeds — is a low-cost, reproducible backbone spurious encoding diagnostic.

Open questions remain: the suppression-vs-dilution mechanistic ambiguity requires directional subspace analysis; the WGA correlation needs per-seed WGA measurements for full confirmation; layer-wise spurious encoding profiles would characterize where GroupDRO's backbone effect is localized; and core attribute probe accuracy would confirm that useful feature encoding is preserved. Two pathways to WGA improvement exist in the Waterbirds WILDS setting; understanding when and why each operates is a prerequisite for principled method selection.

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